

A Technical Introduction to Residual connection neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

Section 1

During the late 1980s, "skip-layer" connections were sometimes used in neural networks. Examples include: Lang and Witbrock (1988) trained a fully connected feedforward network where each layer skip-connects to all subsequent layers, like the later DenseNet (2016). In this work, the residual connection was the form $x \mapsto F(x) + P(x)$, where P is a randomly-initialized projection connection. They termed it a "short-cut connection". An early neural language model used residual connections and named them "direct connections".

Section 2

A residual neural network (also referred to as a residual network or ResNet) is a deep learning architecture in which the layers learn residual functions with reference to the layer inputs. It was developed in 2015 for image recognition, and won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) of that year. As a point of terminology, "residual connection" refers to the specific architectural motif of $x \mapsto f(x) + x$, where f is an arbitrary neural network module. The motif had been used previously (see §History for details). However, the publication of ResNet made it widely popular for feedforward networks, appearing in neural networks that are seemingly unrelated to ResNet. The residual connection stabilizes the training and convergence of deep neural networks with hundreds of layers, and is a common motif in deep neural networks, such as transformer models (e.g., BERT, and GPT models such as ChatGPT), the AlphaGo Zero system, the AlphaStar system, and the AlphaFold system.

Section 3

Previous work Residual connections were noticed in neuroanatomy, such as Lorente de No (1938). McCulloch and Pitts (1943) proposed artificial neural networks and considered those with residual connections. In 1961, Frank Rosenblatt described a three-layer multilayer perceptron (MLP) model with skip connections. The model was referred to as a "cross-coupled system", and the skip connections were forms of cross-coupled connections.

Section 4

Projection connection If the function F is of type $F: \mathbb{R}^n \rightarrow \mathbb{R}^m$ where $n \neq m$, then $F(x) + x$ is undefined. To handle this special case, a projection connection is used:

Section 5

Pre-activation block The pre-activation residual block applies activation functions before applying the residual function F . Formally, the computation of a pre-activation residual block can be written as:

Section 6

Backward propagation The residual learning formulation provides the added benefit of mitigating the vanishing gradient problem to some extent. However, it is crucial to acknowledge that the vanishing gradient issue is not the root cause of the degradation problem, which is tackled through the use of normalization. To observe the effect of residual blocks on backpropagation, consider the partial derivative of a loss function E with respect to some residual block input x . Using the equation above from forward propagation for a later residual block $L >$, we have:

Section 7

Forward propagation If the output of the ℓ -th residual block is the input to the $(\ell + 1)$ -th residual block (assuming no activation function between blocks), then the $(\ell + 1)$ -th input is:

Section 8

$$x_{\ell+2} = F(x_{\ell+1}) + x_{\ell+1} = F(x_{\ell+1}) + F(x_{\ell}) + x_{\ell} \\ \Leftrightarrow F(x_{\ell+1}) = F(x_{\ell+1}) + F(x_{\ell}) + x_{\ell}$$

Section 9

Basic block A basic block is the simplest building block studied in the original ResNet. This block consists of two sequential 3x3 convolutional layers and a residual connection. The input and output dimensions of both layers are equal.

Section 10

where ϕ can be any activation (e.g. ReLU) or normalization (e.g. LayerNorm) operation. This design reduces the number of non-identity mappings between residual blocks, and allows an identity mapping directly from the input to the output. This design was used to train models with 200 to over 1000 layers, and was found to consistently outperform variants where the residual path is not an identity function. The pre-activation

ResNet with 200 layers took 3 weeks to train for ImageNet on 8 GPUs in 2016. Since GPT-2, transformer blocks have been mostly implemented as pre-activation blocks. This is often referred to as "pre-normalization" in the literature of transformer models.

Section 11

Subsequent work Wide Residual Network (2016) found that using more channels and fewer layers than the original ResNet improves performance and GPU-computational efficiency, and that a block with two 3×3 convolutions is superior to other configurations of convolution blocks. DenseNet (2016) connects the output of each layer to the input to each subsequent layer:

Section 12

Stochastic depth is a regularization method that randomly drops a subset of layers and lets the signal propagate through the identity skip connections. Also known as DropPath, this regularizes training for deep models, such as vision transformers. ResNeXt (2017) combines the Inception module with ResNet. Squeeze-and-Excitation Networks (2018) added squeeze-and-excitation (SE) modules to ResNet. An SE module is applied after a convolution, and takes a tensor of shape $R \times H \times W \times C$ (height, width, channels) as input. Each channel is averaged, resulting in a vector of shape $R \times C$. This is then passed through a multilayer perceptron (with an architecture such as linear-ReLU-linear-sigmoid) before it is multiplied with the original tensor. It won the ILSVRC in 2017.

Section 13

During the early days of deep learning, there were attempts to train increasingly deep models. Notable examples included the AlexNet (2012), which had 8 layers, and the VGG-19 (2014), which had 19 layers. However, stacking too many layers led to a steep reduction in training accuracy, known as the "degradation" problem. In theory, adding additional layers to deepen a network should not result in a higher training loss, but this is what happened with VGGNet. If the extra layers can be set as identity mappings, however, then the deeper network would represent the same function as its shallower counterpart. There is some evidence that the optimizer is not able to approach identity mappings for the parameterized layers, and the benefit of residual connections was to allow identity mappings by default. In 2014, the state of the art was training deep neural networks with 20 to 30 layers. The research team for ResNet attempted to train deeper ones by empirically testing various methods for training deeper networks, until they came upon the ResNet architecture.